Project progress check · IF5153 Natural Language Processing

Indonesian **Hoax** Checker



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1,923

hoax items identified and clarified by Komdigi in 2024, about five a day.

Monthly volume ranged from 131 items (February) to 215 (October).

24-25 September 2026

A forged “warning letter” in the Minister’s name claimed that Indonesia’s internet would be shut down for three days. Komdigi called the letter fabricated and told the public there was no reason to withdraw cash or avoid digital transactions.

Delays in identifying and clarifying a hoax can magnify its harm, from public misunderstanding to social unrest

Sources: ANTARA and a UNHAS repository document (2024 counts); Katadata, Bisnis.com and InfoPublik (September 2026)

Project description

We are building an automatic checker for Indonesian news and claims. It takes a claim or article, from one sentence to several paragraphs, formal or informal, and returns a credibility indication, with the official refutation attached when one exists. It works as a pipeline of three NLP tasks in which the output of each becomes the input of the next. It is a first-pass aid that speeds up screening, not the final judge: verdicts with low confidence go to a person, and a claim with no match in the reference base is reported more cautiously.

Language

Indonesian only

Reference base

claims clarified by Komdigi

Status

design stage: no results yet

NLP tasks and owners

1

Named entity recognition

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In raw claim or article text
Out entities and the core claim

2

Hoax classification

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In text plus the entities from task 1
Out hoax or fact label with a confidence score

3

Stance and similarity detection

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In claim, entities, label and score
Out best reference claim, stance label and similarity score

Worked example: one claim through the pipeline

Input raw text	**Pemerintah menghentikan seluruh bantuan Makan Bergizi Gratis mulai bulan depan**
After task 1 entities and core claim	**Pemerintah ORG menghentikan seluruh bantuan Makan Bergizi Gratis CLAIM mulai bulan depan TIME**
After task 2 hoax or fact, with a score	**Pemerintah ORG menghentikan seluruh bantuan Makan Bergizi Gratis CLAIM mulai bulan depan TIME** **Suspected hoax** · with a confidence score from the classifier
After task 3 evidence and stance	**Pemerintah ORG menghentikan seluruh bantuan Makan Bergizi Gratis CLAIM mulai bulan depan TIME** Closest refuted claim in the Komdigi base · stance: supports → **low credibility, refutation attached**

Illustrative example from the design report; no score is shown because there are no results yet.

Data

What each component needs, where it comes from, and what we still have to build.

Datasets by component

Component	Dataset and source	Role in the pipeline	Size
Refuted-claims base	**Indonesian Hoax News Dataset** Kaggle · ireddragonicy	Reference claims, with their clarification text, searched by task 3	About 15,000 records, per the uploader; to verify
NER training data	**NERGrit and NERP, train splits** GitHub · IndoNLP/indonlu	Train and evaluate task 1 on entities; CLAIM and TIME need our own annotation	1,672 and 6,720 sentences, counted
Hoax classifier data	**Indonesian Hoax News Detection Dataset** Mendeley Data · p3hfgr5j3m	Train and evaluate task 2: hoax or valid	600 articles: 228 hoax, 372 valid, per the dataset record
Stance training data	**IndoNLI, train split** GitHub · ir-nlp-csui/indonli	Train task 3: entailment, neutral and contradiction mapped to supports, unrelated and refutes	10,330 pairs, counted

Counted: computed by us from the raw files on 19 and 20 October 2026. The other figures are as stated on the dataset pages and will be re-checked after download.

Refuted-claims base: Indonesian Hoax News Dataset

Hoax clarification articles from the Komdigi portal

The dataset is published on Kaggle by ireddragonicy. Its description says it holds about 15,000 records taken from the hoax clarifications that the Ministry of Communication and Informatics publishes on komdigi.go.id.

We have not yet counted the records or checked the fields. Both are to be verified after download.

Field, as described	Planned use
Cleaned text	The reference claim and its clarification, searched by similarity detection
Raw HTML	Kept for traceability, not used as input
Date	Split by time, and record the retrieval date
Category and tags	Report results per category; check entity overlap
View count	Not used

Field names are as described on the dataset page; column names are to be confirmed after download.

Before use Strip the HTML, drop duplicate and empty records, and check how each article states the claim and the clarification.

NER training data: NERGrit, NERP and our target

Berdasarkan	SK	Ka	Kanwil	Depdikbud	DKI	Jakarta	Nomor	:	7	.
O	O	O	B-ORGANISATION	I-ORGANISATION	B-PLACE	I-PLACE	O	O	O	O

NERGrit, train: 1,672 sentences, 5,321 entity spans (PLACE 2,688, PERSON 1,618, ORGANISATION 1,015). Cased text.

mudik	lebaran	di	indonesia	selalu	identik	dengan	kemacetan	yang	menggila	.
B-EVT	I-EVT	O	B-PLC	O	O	O	O	O	O	O

NERP, train: 6,720 sentences, 13,853 entity spans (PPL 4,845, PLC 4,379, IND 3,065, EVT 1,022, FNB 542). Lower-cased.

Pemerintah	menghentikan	seluruh	bantuan	Makan	Bergizi	Gratis	mulai	bulan	depan
B-ORG	B-CLAIM	I-CLAIM	I-CLAIM	I-CLAIM	I-CLAIM	I-CLAIM	O	B-TIME	I-TIME

Our target annotation on the report's example claim: it adds CLAIM and TIME, which neither corpus has.

Neither corpus labels claims or dates.

Example sentences are the first train sentences of 8 to 11 tokens with two or more entity spans of two types; counts computed by us from the IndoNLU files (accessed 19 October 2026). Our CLAIM span is our reading of the report's example.

Classifier and stance data: Mendeley and IndoNLI

600

news articles in the Indonesian Hoax News Detection Dataset (Mendeley Data): 228 hoax and 372 valid.

What it means for us

The labelled hoax-or-valid data for task 2. It is small and uneven (38% hoax), so we plan a stratified split or cross-validation and will report each class separately. Its items are full articles, while the system may receive a short claim, so we will test on short claims too.

10,330

premise and hypothesis pairs in the IndoNLI training split: 3,476 entailment, 3,415 neutral and 3,439 contradiction (counted by us).

What it means for us

Balanced across the three labels. We plan to map entailment to supports, contradiction to refutes and neutral to unrelated. Its pairs are general sentences (premises of about 21 words, hypotheses of about 8), unlike a claim set against a clarification, so we will test on pairs we annotate.

Experiment design

The questions we test, how the data is split, and the metrics that matter for each task.

Research questions and evaluation protocol

RQ1 NER	Does normalising and augmenting informal text improve entity and core-claim extraction?
RQ2 Classification	Do entity features improve hoax classification over the text alone?
RQ3 Similarity	Does narrowing candidates by shared entities help retrieve the right refuted claim?
RQ4 Stance	Does stance detection separate a repeated hoax from its correction, where similarity alone cannot?
RQ5 End to end	Does the full pipeline with a confidence threshold give a safer indication than a classifier alone?

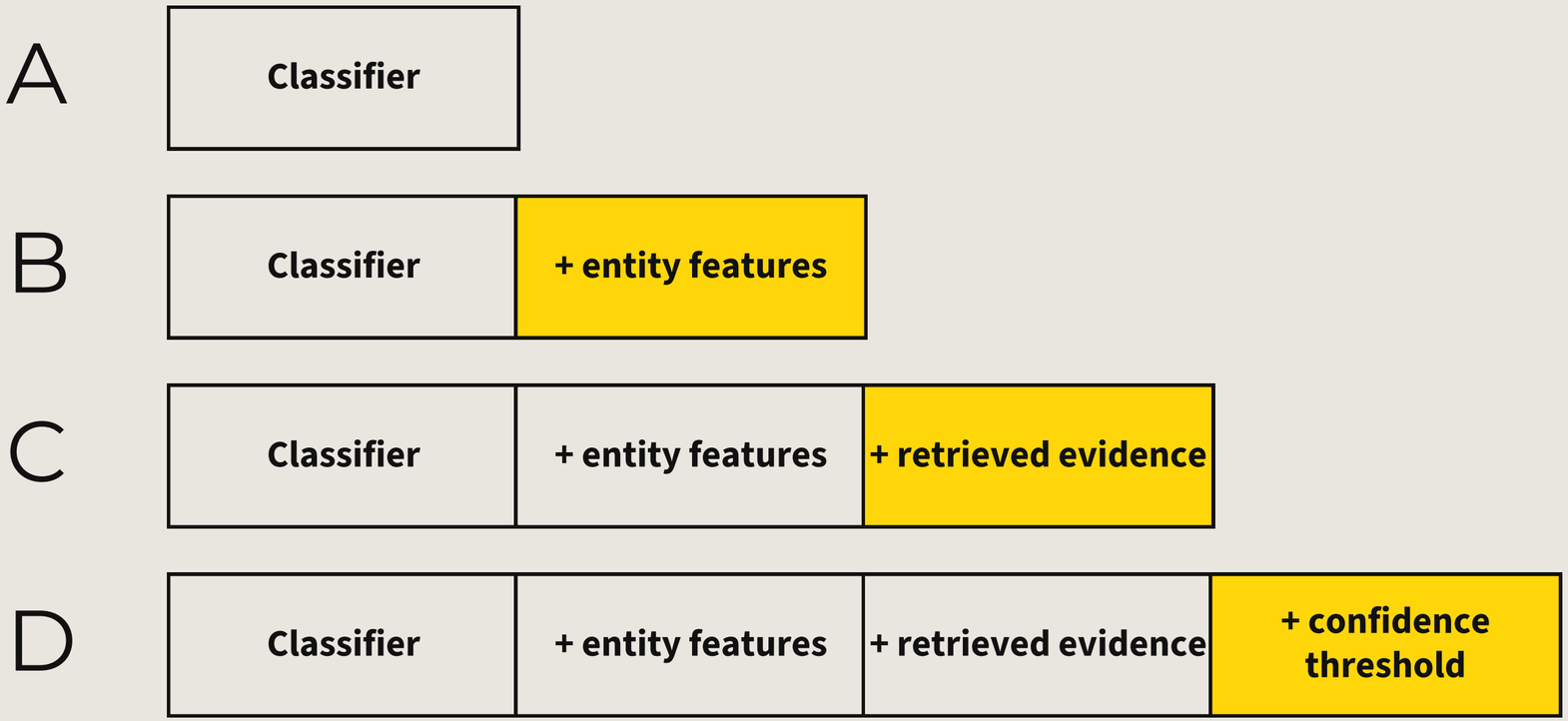
Split by time, not at random. Train and tune on earlier clarifications, test on later ones. Near-duplicates of a claim stay in the same split. **One frozen test set.** Every ablation is scored on the same held-out data. **Repeated runs.** Three seeds per model; we report the mean and the spread.

Experiments per task

Task	Baseline	Stronger model	Metrics	Ablation
Named entity recognition 13523139	BiLSTM-CRF tagger	IndoBERT token classification	Entity-level F1, overall and per type; CLAIM F1; formal versus informal text	With and without text normalisation and informal augmentation
Hoax classification 18223124	TF-IDF with logistic regression	IndoBERT sequence classifier	Macro-F1; hoax-class precision and recall; calibration of the confidence score	With and without NER entity features
Similarity detection 13523136	BM25 retrieval	Sentence embeddings with cosine similarity	Recall@k and MRR of the right refuted claim	With and without entity-based candidate narrowing
Stance detection 13523136	Similarity threshold only	NLI model fine-tuned on claim and clarification pairs	Macro-F1 over supports, refutes, unrelated; support-versus-refute confusion	Similarity alone versus similarity plus stance

Proposed design. IndoBERT is the model family released with IndoNLU (Wilie et al., 2020).

End-to-end evaluation



Pipeline variants, each adding one element to the previous one. Yellow marks what is new.

Metric for each risk in the BEAM figure

Entity extraction error
entity F1 per type, formal versus informal

False positive, false negative
error rates per class and per category

Stance detection error
support-versus-refute confusion

Outdated refutation base
retrieval recall as claim age grows

Review load (threshold)
share held for review versus accuracy of the rest

A good result is fewer missed hoaxes at an acceptable review load.

Status and open questions

Next

1. Collect and clean the Komdigi list into the refuted-claims base
2. Annotate a seed set with entity and claim labels, and choose the fact class
3. Train a baseline for each task on a time-based split
4. Integrate the pipeline and run the ablations
5. Analyse errors, choose the confidence threshold, write the final report

Done project design report (description, three NLP tasks with owners, BEAM notation with risk layer, system description).

Questions for feedback

Which source gives a fair “fact” class for the classifier?

Is a CLAIM span the right annotation target, or should we label claim sentences?

Is a time-based split enough to stop recurring hoaxes from leaking?

The system returns a credibility indication with its evidence attached, and passes low-confidence cases to a person.

Entities feed both the classifier and the evidence search.

Public NER corpora label neither claims nor dates; the knowledge base must be kept fresh.

Each risk in the BEAM figure has a metric in the experiment design.

Next: build the refuted-claims base, annotate a seed set, train baselines.